SP PROJECT REPORT

Q1

Part a)
Given Non uniform jittered sampling how would you go about reconstructing the signal?

Our solution:

Sinc would not work as ideally as it normally does in this case because in uniform sampling we use the expression sinc((t-nTs)/Ts)
At t=mTs we can see Ts cancels in the numerator an denominator and we get sinc of an integer which is 0. Thus only t=nTs contributes and we get perfect reonctsruction.

But here because t-=mTs+k*delta, the Ts wont cancel in the numerator and denominator and thus other terms contribute leading to imperfect reconstruction.

Thus we will be using spline which is a cubic interpolation. It guarantees continuity of first and second derivative and thus the curve looks smooth.
The idea is that given the non uniform samples, it creates a curve with all the points containing the non uniform samples and then evaluate this curve at the ideal samples.

We will be measuring its performance with respect to jitter factor.

Upon simulating we find that as the jitter facor increases reconstruction accuracy decreases.
This is because the samples are further from their dieal positions.

Q1 b)

Problem Statement: Some samples are missing with probability ‘p’. Find the accuracy of reconstruction wrt p.

Our Solution:
First fill out the missing samples using average of neighbours, then using sinc interpolation reconstruct the entire signal.

We observe that reconstruction degrades as p increases. This is because, more samples are missing and we thus have lesser information.
This method would produce quite faulty reconstruction if the probability was high. However it was given in the qs that we must measure from 0-0.1 thus, this method works quite well for the given constraints.
Another potential issue could be due to linear interpolation not being smooth, it adds high frequency components to the signal thus distorting its band limited nature.

However the advantage of this method is we get more information and thus lower error compared to directly doing sinc interpolation

Q2

Problem Statement:
Given files containing recordings of drum sounds, identify when a hit happens, the duration of the hit and the various instruments which are hit.

Our solution:
First we split the signal into small strips in the time domain. You perform short time fourier transform and get a spectrogram which is frequency x time.Then we find the energies corresponding to each strip of time by summing up all frequencies using the appropriate formula. First we smoothen out the signal using low pass filter so the noise is not misidentified as a ‘hit’. We then normalise the energy values with respect to the highest energy value so the analysis becomes simpler.You then set a threshold value of energy for what we consider a hit and the duration of the hit is time taken to go from that threshold value down to a certain %(10) below it.

Now we will identify the various instruments by analysing the characteristics of the signal in frequency domain. The parameters we analyse are:

1) Centroid: It can be thought of as a ‘Center of Mass’ with respect to frequency.
2) Bandwidth: How wide the energy is distributed across frequencies
3) Spectral Roll Off: The frequency at which total energy becomes 85% of the total energy. A low value of spectral roll of indicates high energy components in low frequency and vise versa.
4) Log of Total Energy: measures overall loudness/intensity
5) High Frequency/Low Frequency: We set a threshold value off frequency and count how many frequency components are above and below it and divide them. This tells us if there’s many high frequency components in the signal etc.

K mean:
We have all the data points of all these characteristics we have mentioned now, we now wanna group the similar ones.

So lets say we wanna group 10 data points into 3 clusters based on closeness.
First pick 3 random points
Compute distance of each data point from all centroids and map the data points to the centroid which is at minimum distance from it.
Then find the mean of each cluster(which is the new centroid).
Recursively do it again and again until the cluster members dont change.